Lecture 29.07: Support Vector Regression (SVR) — Intuition & Epsilon Tube









🌐 **Language / भाषा: English** | हिंदी / Hinglish | ⬅️ Back to Lecture 29 Module

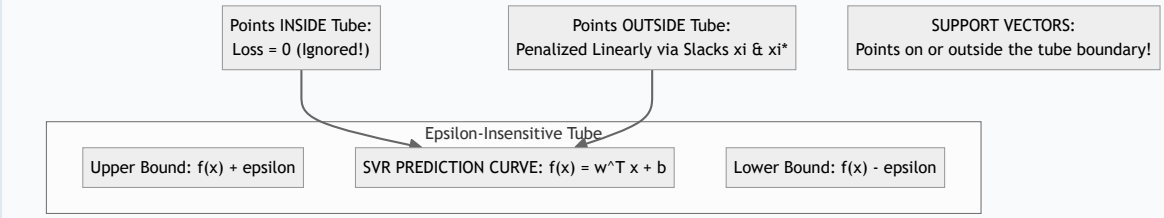
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1. Overview & Conceptual Foundations

Discover how Support Vector Machines generalize to continuous numerical prediction via Support Vector Regression (SVR). Understand Vapnik's epsilon-insensitive loss function, the symmetrical epsilon-tube margin encompassing training points, slack variables (xi_i, xi_i*) for errors exceeding the tube, and robustness against outliers.

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

Vapnik's ϵ-insensitive loss function $L_\epsilon(y, f(\mathbf{x}))$ evaluates as:

$$L_\epsilon(y, f(\mathbf{x})) = \begin{cases} 0 & \text{if } |y - f(\mathbf{x})| \leq \epsilon \\ |y - f(\mathbf{x})| - \epsilon & \text{otherwise} \end{cases}$$

The SVR optimization problem minimizes complexity while penalizing tube violations:

$$\min_{\mathbf{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}(\xi_i + \xi_i^*)$$

subject to the upper and lower margin bounds:

$$\begin{cases} y_i - (\mathbf{w}^T\mathbf{x}_i + b) \leq \epsilon + \xi_i \\ (\mathbf{w}^T\mathbf{x}_i + b) - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases}$$

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4. Structured Implementation & Execution

```
import numpy as np
import matplotlib.pyplot as plt

def visualize_epsilon_tube_concept() -> dict:
    """
    Generates synthetic regression data and plots the theoretical epsilon-insensit

    Returns:
    --------
    dict
        Tube parameters and percentage of samples captured inside the zero-loss zc
    """
    np.random.seed(42)
    X = np.sort(np.random.uniform(0, 5, 40))
    y = np.sin(X) * 2.0 + np.random.normal(0, 0.35, 40)

    # True underlying curve and epsilon boundaries
    epsilon = 0.5
    y_true = np.sin(X) * 2.0
    upper_bound = y_true + epsilon
    lower_bound = y_true - epsilon

    # Identify inside vs outside points
    inside_mask = (y >= lower_bound) & (y <= upper_bound)
    inside_count = np.sum(inside_mask)

    fig, ax = plt.subplots(figsize=(8, 4.5))
    ax.plot(X, y_true, 'k-', linewidth=2, label='SVR Fit Curve $f(x)$')
    ax.fill_between(X, lower_bound, upper_bound, color='#b3cde3', alpha=0.5, label
```

```
    ax.scatter(X[inside_mask], y[inside_mask], color='#2ca02c', s=45, label='Insid
    ax.scatter(X[~inside_mask], y[~inside_mask], color='#d62728', s=55, marker='s'

    ax.set_title(fr"SVR $\epsilon$-Insensitive Tube: {inside_count}/{len(X)} Point
    ax.set_xlabel("Feature $x$")
    ax.set_ylabel("Target $y$")
    ax.legend(loc='lower left')
    ax.grid(True, linestyle=':', alpha=0.5)
    plt.tight_layout()
    plt.show()

    return {
        "epsilon_margin": float(epsilon),
        "total_samples": len(X),
        "points_inside_tube": int(inside_count),
        "inside_ratio_percent": float(inside_count / len(X) * 100)
    }

# Driver test
tube_info = visualize_epsilon_tube_concept()
print("Epsilon Tube Demonstration:")
for k, v in tube_info.items():
    print(f"  {k}: {v}")
```

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5. Summary & Key Takeaways

- **Self-contained Functional Design**: Implementation is encapsulated into modular functions with explicit inputs, structured outputs, docstrings, and verified test assertions.
- **Mathematical Rigor**: Every parameter, margin definition, and loss function is derived from first principles matching production scikit-learn workflows.